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RELEVANT WORK EXPERIENCE/INTERNSHIPS

GOLDMAN SACHS *Associate Software Engineer* **Apr 2025 – Present**

- Optimized distributed systems utilizing Java and Akka for actor-based concurrency, enhancing data pipeline throughput and reducing processing latency while managing hardware limitations.
- Architected and managed high-volume data streaming pipelines leveraging Kafka and RabbitMQ, significantly improving system scalability and reliability for handling 10k+ transactions/messages daily.
- Led Site Reliability Engineering initiatives for the platform by developing React + REST API dashboards for users to define and monitor SLOs/SLIs, enhancing system observability and uptime.
- Led AI initiatives by integrating an agentic AI workflow assistant and writing skills that eliminated manual support triage time by identifying false positives and suggesting actionable fixes.
- Built and maintained backend services in a proprietary object-oriented language to support finance workflows and regulatory requirements, while onboarding multiple trading desks by delivering new features, managing resource utilization, and coordinating closely with cross-functional stakeholders.

KULICKE & SOFFA *Advanced Software (Systems) Engineer* **Jul 2022 – Apr 2025**

- Built real-time C++ systems on VxWorks and Linux over hypervisor for semiconductor equipment, designing multi-threaded data pipelines and integrating telemetry streams with monitoring services.
- Owned end-to-end delivery of Vertical Wire Height Monitor, building Qt/C++ applications (MVC, 10+ UI screens, custom X11 rendering) integrated with real-time wire visualization, hardware control, SQLite database.
- Designed and implemented logging and telemetry export capabilities across 10+ features, such as vision data, reducing debugging time by 80% for both customers and in-house engineers.
- Extended the real-time process-monitoring pipeline by adding new signals and dashboards using FlatBuffers, SQLite, and Python analytics services, improving error detection coverage.
- Developed automated calibration tools using computer vision and mathematical modeling to compensate for temperature-induced drift, reducing manual calibration time by 30%.
- Maintained code quality using cppcheck, PVS-Studio, and clang-tidy static analysis tools. Managed cross-platform builds (Windows/Linux) with CMake and collaborated via Git, Jira, and Confluence in an agile environment across the global teams.

GWS *Machine Learning Engineer Intern* **May 2022 – Jul 2022**

- Led a 3-person engineering team to develop an end-to-end audio biodiversity monitoring system, utilizing Python and TensorFlow to automate the identification of 4 distinct poultry species in noisy environments.
- Implemented unsupervised clustering algorithms (K-Means, DBSCAN, GMM), utilizing the Elbow Method and BIC scores to optimize hyper parameters Architected a semi-supervised learning framework combining a Convolutional Neural Network (CNN) for feature extraction with Gaussian Mixture Models (GMM).
- Facilitated data-driven decision making by translating technical findings into actionable insights in stakeholder presentations, communicating model performance metrics (Confusion Matrices, Log Loss) to guide project strategy

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, Typescript, Python, R, C, C++, Dart, SQL, HTML, CSS, Shell Scripting
Frameworks & Libraries: React, React Native, Flutter, NextJS, NodeJS, MaterialUI, REST API, Figma, Balsamiq, Robot, LabView
Databases & Tools: MongoDB, NoSQL, MySQL, PostgreSQL, Hadoop, Spark, Docker, Unix, Linux, Git, Kafka, RabbitMQ, Claude, Copilot
Machine Learning & Data Analytics Tools: Pandas, Numpy, Seaborn, Tableau, Streamlit, Scikit-Learn, Tensorflow, PyTorch, PowerBI

PROJECTS & HACKATHONS

Space Together (Hack&Roll) **May 2021**

- Developed a mobile application in Swift in tandem with CoreBluetooth and BlueTrace that alerted users with a fun customizable ringtone when they were too close to other people. This was intended to encourage and make safe distancing fun.

NapMap (Project Orbital) **May 2019**

- Used React Native and Google API to develop a mobile application that used geolocation to alert users that they were reaching their destination.

EDUCATION

National University of Singapore (NUS) **Aug 2018 - May 2022**
Bachelor of Computing (Computer Science), Honours

ADDITIONAL INFORMATION

- Fluent in English and Mandarin (both spoken and written)
- Touch Rugby Club and Varsity Player, ex-National and Varsity International Chess Player